FYP 2 Progress Report — Fly.io Shadow Deployment (2026-10-11, Week 2 start)

Project: Apa News Hari Tok? (Telegram news bot for Sarawak local news) **Author:** Vincent Lee Cheng Hong (BCS24020018) **Period covered:** 2026-10-11 (FYP 2, Week 2 start) **Status:** Bot redeployed to Fly.io running current `main` in utility-alert **shadow mode**; first server-side shadow polls verified. Main RSS prefetch partially blocked by datacenter-IP filtering (documented below). **Code:** `main` @ `176962e` deployed as Fly release v111

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1. Why Fly.io (owner decision, 2026-10-11)

- Owner chose to keep the deployment on **Fly.io** (`apanewsharitok`, region `sin`) rather than Oracle Cloud "for now". Oracle container + GitHub-push auto-deploy remain paused and available as fallback.
- Driver: continuous multi-day shadow-log accumulation (Stage 1b of `04-utility-alerts.md`) without keeping the laptop on 24/7.
- **Ollama is not required** for this stage: the utility-alert classifier is rule-based and runs with Ollama down (04 §3.3, project rule #8). The Ollama cloud quota reset (due ~2026-10-12) only matters for optional AI enrichment.

2. Deployment actions

Step	Result
flyctl v0.4.115 installed (workspace `tools/flyctl/`, portable)	[OK]
Auth: session had expired; owner re-authenticated (`bcs24020018@student.uts.edu.my`)	[OK]
`.dockerignore` hardened: added `*.session`, `tools`, `logs`, `*.pdf`	[OK] — the Telegram session file was previously baked into image layers; no longer
`flyctl deploy --app apanewsharitok --remote-only`	[OK] release v111 (previous image was from 10-06 09:22, predated Phase 4)
Secrets	Reused existing set: `TELEGRAM_BOT_TOKEN`, `DATABASE_URL` (Supabase pooler), `TELEGRAM_SESSION_STRING`, `TELEGRAM_API_ID/HASH`, source config. No `UTILITY_ALERT_*` overrides → code defaults apply (`SHADOW_MODE=true`, 10-min poll, 24h lookback, 3/h cap)
Machines	**Scaled 2 → 1.** Both machines polled Telegram with one token → `Conflict: terminated by other getUpdates request`. A polling bot must be single-instance; scaling to 1 also halves cost. (Prior 2-machine config predates this understanding.)

3. Shadow-mode evidence (live, 2026-10-11 UTC)

```
09:34:06Z [utility-alert] poll in 15.8s: {'fetched': 8, 'classified': 6, 'new': 0,
'duplicates': 6, 'rejected': 2, 'sources': {'telegram': 0, 'jbalb': 6, 'news': 2}}
09:44:06Z [utility-alert] poll in 15.7s: {'fetched': 8, 'classified': 6, 'new': 0,
'duplicates': 6, 'rejected': 2, 'sources': {'telegram': 0, 'jbalb': 6, 'news': 2}}
```

- **Idempotency from a datacenter IP confirmed**: both polls stored 0 new rows and classified all 6 JBALB notices as duplicates of the 10-06 shadow run — cross-instance dedup via `source_ref` + title-Jaccard works against the shared Supabase database.
- **Zero pushes**: shadow mode only logs would-be deliveries.
- **Telegram source returned 0 twice with no auth errors** in the logs. Either no `swbnews` posts fell in the 24h lookback, or the session is silently struggling from Fly's IP. **Watch item** (see §5).
- No `Conflict` errors since the second machine was destroyed (09:29Z).

4. New finding: news-site anti-bot blocking from datacenter IPs

The regular RSS prefetch (separate from utility alerts) gets hard-blocked from Fly's IP range:

```
09:43:09Z Error fetching feed https://news.seehua.com/feed/: 403 Forbidden
09:43:09Z Error fetching feed https://www.theborneopost.com/feed/: 403 Forbidden
```

```
09:43:14Z Error fetching feed https://www.freemalaysiatoday.com/.../feed/: 403 Forbidden
09:43:16Z Error fetching feed https://rss.app/feeds/EYJxxljrFQtbWR8K.xml: 402 Payment Required
09:43:21Z Error fetching feed https://www.tvsarawak.my/feed/: 403 Forbidden
```

- See Hua / Borneo Post / FMT / TVSarawak block Fly.io IPs (Cloudflare-class bot protection). The utility-alert **Google News RSS watch is unaffected** (works from the same IP), so the alert pipeline is intact.
- The expired rss.app subscription (402) is pre-existing and unrelated to Fly.
- **Implication**: the main evening-digest article pipeline will starve on these sources while hosted on Fly. Options: (a) prefetch from the laptop / another residential-IP host, (b) accept reduced coverage and document it as an anti-bot limitation (strengthens thesis §5.4.5 source-coverage argument), (c) proxy. Deferred to owner decision — does not block Stage 1b.

5. Stage 1b acceptance-criteria tracking (from 04 §5)

Criterion	Status 2026-10-11
Precision ≥ 90% (sample ≥ 20 stored alerts)	Pending — needs multi-day accumulation; sample still the 10-06 set
Zero pushes	[OK] Verified (two polls, shadow mode on)
Quiet hours (00:00–06:00 held to 06:00 Asia/Kuching)	Pending — no overnight alert yet
Rate cap ≤ 3/h per user	Pending — no live alert yet
Stability (10-min ticks, non-fatal per-source errors)	[OK] Two consecutive on-schedule polls; per-source failure isolation verified 10-06
No cross-poll duplicates	[OK] Verified across laptop → Fly instance boundary (stronger test than same-host)

6. Open items / next

1. **Telegram source watch**: if `swbnews` posts within the next 24–48h and `telegram` count stays 0, investigate the session from Fly's IP before Stage 2 sign-off.
2. **Password rotation** (Supabase credential exposed 10-05) — still open; now also matters because the Fly app holds `DATABASE_URL`.
3. **Log durability RESOLVED 2026-10-11 (commit `855f2f8`)**: every poll now writes a summary row and every first-due would-be delivery writes a deduped row to the new `utility_alert_shadow_log` table (Postgres/SQLite, gated by `UTILITY_ALERT_SHADOW_LOG`, shadow-mode only, retention-purged). Evidence survives log-stream rotation; the DB is the Stage 1b archive. Remaining: redeploy to Fly so the running instance ships evidence (the v111 image predates this change).
4. Owner decision: RSS 403 mitigation strategy (§4).
5. Local laptop bot must stay **off** while the Fly instance runs (same-token conflict).

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